

Sentiment Analysis on YouTube Videos for kids: Review

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Abstract—With the increasing emergence of social network applications, people now use social media to share their opinions on daily issues and analyse people's interactions and reactions toward daily activities such as buying a product, watching videos, community services and other viral situations. A significant number of children watch videos using web platforms such as YouTube. However, such platforms can contain useful or harmful videos to children. Many attempts have been conducted in order to rank the quality of YouTube Videos based on the sentiment analysis approach. However, few attention has been given to Kids' YouTube videos. Therefore, this paper investigates existing methods and techniques that concentrate on YouTube video sentiment analysis. The methods classify them based on machine, deep learning approaches. This paper can be beneficial to many researchers in data mining, sentiment analysis area and more related research areas.

Keywords—YouTube; Sentiment Analysis (SA); Machine Learning (ML); Deep learning (DL).

I. INTRODUCTION

Currently, social media applications are involved in our lives in one way or another, and users are expressing their opinions and emotions on its different platforms constantly in the form of microblogging (e.g., YouTube, Twitter, Facebook, etc.), added comments and posted content through different accounts. Such information can be utilised to make relevant recommendations and be useful to the user on a daily basis. Due to this, researches based on users' feelings gained wide attention lately using Sentiment Analysis.

Sentiment Analysis (SA) is the analysis of the attitude of a speaker or a writer with respect to a certain topic; it includes a variety of Natural Language Processing (NLP) tasks like polarity and degree of sentiment [1]. Importantly, it has been used as practical techniques in analysing user comments, opinions and movie reviews to extract the sentiment of a text and classify it as positive and negative or natural. That includes feature extraction from users' opinion, and it plays an important role in our daily decision-making about a product or a movie [2,4,5].

SA has become a main research field in the NLP community; therefore, it sports a wide spectrum of practical applications that include opinion mining and emotion extraction in social media trend predictions [3,6].

YouTube is a platform for sharing videos and allows to users upload, view and share videos [7]. In addition, several videos allow users to leave comments. YouTube includes movies, animation and educational videos.

YouTube ranks video content based on the number of likes, dislikes and comments [8-9-10]. Also, most users search for a video and the associated comments before watching the content. So, user comments tend to provide feedback about the quality of the YouTube videos. Moreover, analysing and classifying user comments, opinions and movie reviews into positive and negative based SA is very important for YouTube [11]. Technology has changed the way people receive and find information, and the inclusion of social media platforms has made it possible to develop user-generated data. The utilisation of these platforms has enhanced the effectiveness of annotating and publishing content significantly through different accounts [12]. The effectiveness of social media platforms has been enhanced for both multimedia organizations and professional content developers [12]. Many researchers proposed methods to detect irrelevant videos based on users' comments, spam and promoters' accounts [13,14].

The existing researchers did not utilise likes, dislikes and comment on YouTube videos to suggest videos for users to watch. In this study, these parameters play an important role in helping find the proper YouTube video for the user. Using polarity and degree of sentiment are NLP tasks that guide the users in a convenient way to select the likable YouTube videos. Also, analysing comments as positive and negative allows users to choose a suitable category to watch an instance for positive and negative comments on YouTube videos as shown in Fig.1.



Fig. 1. Example of Positive and Negative Comments on YouTube Videos.

YouTube is one of the platforms targeted for children due to its popularity. Such platforms can contain both useful and harmful videos to their demographic. Notably, YouTube introduced the YouTube Kids application¹, which enables parents the ability to control what their children should watch on YouTube. In spite of YouTube's efforts to reduce the spreading of videos unsuitable to children, the disturbing videos still appear for them [13,15,16], and it is essential to ensure that they access suitable content [17]. There are studies that have given attention to this issue to find out whether or not the video is appropriate on YouTube, but the issue continues to exist and requires a more precise solution and improved framework.

Therefore, this paper provides a review summary on the existing approaches that were utilised, methods that concentrated on SA or ranking the YouTube videos based on video comments, title and the number of likes through sentiment analysis to detect the suitability of YouTube videos and provide safety for children.

The structure of the paper is organized as follows. Section II presents various studies about social media sentiment analysis using supervised machine learning approaches. Section III presents different related studies about social media sentiment analysis using supervised deep learning approaches. Section IV discusses recent studies of social media sentiment using ensemble approaches. Section V offers other research focusing on social media sentiment using other approaches. Finally, a conclusion will be covered in Section VI.

II. SUPERVISED MACHINE LEARNING APPROACHES

This section presents studies utilising supervised machine learning approaches for social media sentiment analysis. Such approaches have been applied to the sentiment analysis problem. This mainly can be solved using supervised classification techniques and, in particular, text classification techniques. Therefore, it is called supervised learning. The classification tasks based on machine learning algorithms require two sets of data: training and testing. A training set is used by an automatic classifier to learn the distinguishing properties of documents, and a test set is used to validate the automated classifier's performance. A list of supervised machine learning algorithms includes K- Nearest Neighbour (KNN), Support Vector Machines (SVM), Random Forest (RF), Naive Bayes (NB) and Logistic Regression (LR). Random Sample Consensus (RANSAC) and Decision Trees (DT) for text classification.

Tanesab *et al.* [2] conducted a sentiment analysis of YouTube comments using the SVM method. There were 1000 text comments written in Indonesian used as a dataset. Additionally, the collected dataset is related to the performance of Ahok as a governor. In their search, sentiment analysis has been used to recognize positive and negative sentiments. So, they used Lexicon Based sentiment analysis to detect positive and negative comments. The performance of the SVM was evaluated using the confusion matrix to collect the weighting percentage achieved using SVM. As a result, their model achieved 84% accuracy, 91% precision and 80% recall.

Abd El-Jawad *et al.* [4] introduced a hybrid system for sentiment classification that used neural networks and text mining. Moreover, the performance of deep learning algorithms and different machine learning techniques were evaluated and compared. The proposed methodology comprises the following steps. First, the data collection phase where more than 1 million tweets were collected from the Kaggle repository, GitHub and UCI archive of the English language (from Twitter 140 API). Second, the pre-processing step where all noise, duplicate tweets, stop words and punctuation symbols are removed. Third, the feature extraction and classification step in which tweets were examined to decide on particular features and represent every tweet by a vector that the classifier can interpret. Finally, the evaluation step is where the tweets are classified using a number of classification algorithms including NB, RNN, DT, NN, RF and a hybrid model. It was trained using 75% of the dataset, and 25% of the residual dataset was tested. Results indicate that the highest accuracy rate is equal to 83.7%. Additionally, 87.1% resulted in sensitivity and 79.3% in specificity. So, this illustrates that the efficiency of the system's hybrid learning approach outperformed that of standard supervised approaches.

Khan *et al.* [13] applied supervised machine learning techniques for sentiment classification of user reviews with a comparative opinion mining perspective. In their study, they manually gathered business-related comparative reviews from YouTube, annotated review sentences into positive, negative and neutral classes, and finally stored 6,000 reviews into a machine-readable CSV file. Moreover, a number of pre-processing methods were applied to the dataset including tokenization, tagging, spell correction, special characters and hashtags removal. Then, they classified comparative reviews into nine different classes and then divided the dataset into 80% for training and 20% for testing respectively. Additionally, they used training and evaluation supervised ML models, such as NB, SVM, KNN, DT, Gradient Boosting, LR and RF classifier, using the Anaconda framework. In summary, they applied all these classifiers on the comparative reviews dataset in order to evaluate the performance and efficiency of each classifier. They evaluated all the classifiers with accuracy, precision, recall and F-Score for 6,000-review datasets (i.e., iOS vs Android) and another 3,000-review datasets (i.e., Microsoft vs Google). The result shows that the RF classifier performs best on the 6,000-review dataset, but the DT outperforms on the 3,000-review dataset.

Muhammad *et al.* [19] have proposed an approach to detect sentiments present in YouTube videos using the combined method. NB and SVM classifiers were combined as a hybrid machine learning called NBSVM. However, the performance differs in all features, variants and numbers of data collection. The combination of NB with SVM achieved a better accuracy level. NBSVM classifier achieved an 87% F1-score, 91% precision score and 83% recall, considering the highest score obtained.

Amanda and Negara [18] have conducted experiments to analyse YouTube data by implementing ML algorithms for YouTube data classification. The performance of different algorithms was evaluated and compared. The data utilised was taken from the YouTube website. Different data pre-processing methods used the Natural Language Toolkit (NLTK) library to remove all punctuation, convert letters

¹ <https://www.youtube.com/kids/>

into lowercase forms, etc. Classification algorithms such as RF, SVM and NB were applied for classification. In the conducted experiment, the RF classifier achieved 82%, SVM also attained 82% and NB achieved 88% on classification accuracy. Consequently, the result of conducted experiment shows that the NB classifier attained the highest accuracy for YouTube data classification.

Novendri *et al.* [9] proposed NB-based sentiment analysis work on YouTube movies' trailers using users' comments. In their study, they used Selenium WebDriver as a dataset crawling technique to acquire user comments from YouTube. The labelling process was performed to label the dataset as positive or negative. After applying tokenization, stemming and filtering on the stop list algorithm on the 998-comment dataset, the features were extracted using TF-IDF. Finally, NB is implemented on the collected dataset for sentiment analysis. Consequently, the performance evaluation results show that NB achieved 81% accuracy, 74.8% precision and 75.2% recall.

Yasin *et al.* [20] introduced a machine learning based on dual language sentiment analysis model for YouTube videos ranking in Asia. YouTube ranking system ranks video content based on their number of likes, dislikes and comments, and most users search videos and read its comment before watching them. They selected the English and Roman Urdu dataset from Kaggle, applied Instant Scrapper on datasets, annotated them as English-positive, English-negative, RU-positive and RU-negative. After data cleaning, Count Vectorization, TF-IDF Vectorization and Word2vec Vectorization were applied as feature selection techniques. After that, a comparative analysis is performed with LR, RF, Linear Regressor and RANSAC machine learning models where outperform others and Logistic Regression achieve 87% accuracy with 84% precision and 91% recall.

III. DEEP LEARNING APPROACHES

The section explains the most popular approaches of deep learning algorithms that have been used for better results in classification tasks. They have also been used to perform sentiment analysis on data (i.e., YouTube, Twitter, Facebook, etc.). Most use pre-determined algorithms like Recurrent Neural Networks (RNN), Convolution Neural Networks (CNN) and backpropagation algorithms. In [19,21] focused on detecting unsafe content for children and its promoters on YouTube. CNN-based deep learning models were employed on video frames to detect unsafe content promoters. Moreover, they used three types of metadata features based on video, comments and uploaders. The experimental results have shown that the neural network model achieved 85.7% detection accuracy as it provides build a safe YouTube.

Ramadhani and Goo's [22] deep-feed neural network models were used for Twitter sentiment analysis. The proposed deep feed neural network consists of three hidden layers, and it attained better results of average accuracy of 76.24% compared to the result of the Multilayer Perceptron (MLP), which gave an average accuracy of 60%.

Ali *et al.* [23] introduced a sentiment analysis classification model based on deep learning methods. Moreover, the proposed model combined Long Short Term Memory networks (LSTM) and CNN. The proposed CNN-

LSTM is compared with LSTM, MLPs and CNN to prove the applicability of the proposed model. The results showed that the deep learning models performed better compared to machine learning like SVM, NB and RNTN, which were published using English datasets in other research. However, the IMDB datasets utilised contain 50k movie reviews, where reviews were divided into negative and positive reviews, each of which equals 50%. Of that, 50% were positive reviews and 50% were negative. The CNN-LSTM model attained 89.2% accuracy outperforming CNN, MLP, and LSTM that achieved accuracies of 87.7%, 86.74% and 86.64 respectively.

Cunha *et al.* [24] presented a deep neural network to analyse sentiments in YouTube video comments written in the Brazilian Portuguese language. There were two videos with 1000 comments each from 'Joice Hasselmann.' Therefore, they classify user comments into three categories about how users feel toward the videos: video quality and YouTuber, how well a video covered the topic, and how relevant the video is. Then, they represent each classification result by neutral, positive and negative. First, they perform lexical pre-processing on comments to remove special characters, then pre-process data to normalis and finally apply a 6-layer deep neural network on data. An embedding layer is used to represent the input as a tensor, followed by two convolutional layers for feature extractions. The fully connected layer is used after the convolution layers for the classification process. They evaluated their model with Mean Squared Error (MSE) and compared classification accuracy with a statistical Bayesian classifier. The results showed a range between 60% to 80% classification accuracy, which outperformed a statistical model.

Postalcioglu and Aktas [25] proposed various neural network models comparison for nostalgic sentiment analysis of YouTube comments. In this study, Multi-Layer Perceptual Neural Network (MLPNN), CNN and LSTM algorithms were used on songs' comments to assess their level of nostalgia to an audience. To find the nostalgic comments, they included comments from old songs with YouTube data API into a CSV file with nostalgic and non-nostalgic labelled. To extract features from Natural Language Processing, they applied the Global Vectors for Word Representation (GLoVe) method. Moreover, they input these words to the Neural Network that embedded words into a lower-dimensional space. Then, the dataset was pre-processed using tokenization, stop word and noise removal. After training models on the dataset, the performance of neural network models was evaluated, where MLPNN show 78%, CNN show 88, and RNN-LSTM shows 88% classification accuracy.

IV. ENSEMBLE APPROACHES

The main objective of ensemble classifiers is to obtain more precise and accurate predictions by combining multiple classifiers. Text classification processes have proven to benefit from the principle of ensemble learning. In the area of social media SA, ensemble methods showed clear advantages over traditional supervised machine learning.

Xia *et al.* [26] studied the purposes of sentiment classification and the impact of utilising ensemble learners for these purposes. The study achieved a more powerful

classifier that combines diverse feature sets and multiple classifiers to achieve efficiency. Word-relations and POS are feature types, and SVM, NB and MaxEnt are classifiers used in this research. Fixed grouping, weighted grouping and meta-classifier grouping are investigated and evaluated as various ensemble classifiers. Furthermore, the production of significant improvements originated from different classifiers with diverse feature sets that are showed from the outcomes obtained using ensemble classifiers.

Martinez-Cámara *et al.* [27] conducted various experiments to examine the effect of information acquired as a statistical feature selection method on the classifier's accuracy. Utilising information gain for reducing feature vector dimensionality before the collected result presented clear improvements in the classification accuracies. The results illustrated that the accuracy score of the majority-voting ensemble classifier is 93.94 compared to the accuracy score of the SVM is 92.71 for sanders datasets. Moreover, the Stanford-1K dataset achieved a 78.70 accuracy score in the majority-voting ensemble classifier and a 78.10 accuracy score in the SVM. They achieved accuracy for NB and the ensemble methods, 85.09 and 84.75 respectively, consecutively for the HCR dataset. The overall results show that ensemble methods have better performance when compared with traditional supervised machine learning methods.

V. OTHER APPROACHES

Several researchers used various miscellaneous techniques for comment analysis on YouTube. Rangaswamy *et al.* [28] proposed a technique to extract information by using the video URL such as author, title, genre, keywords, etc. The metadata was analysed to make a decision based on comments for determining the sentiment of each video whether the given video was positive, negative or neutral. Nevertheless, the size of datasets was too small to use in a comprehensive study.

Bhuiyan *et al.* [29] proposed a retrieve method that is an NLP-based method for the most relevant YouTube video based on sentiment analysis and video rating. The main issue addressed in the study is that low-quality and irrelevant videos are ranked at the top of the results of searching. The proposed method comprises four steps. First is the comment collection and pre-processing step where all the stop words and time, numbers, punctuations, and others language except English are removed. Second, two datasets are generated for each video, the first dataset for words after removing stopwords and the second dataset contains all the adjectives. Third, they applied the SentiStrength sentiment classifier that shows the estimates for both the strength of positive and negative sentiments. Finally, the standard deviation value is calculated for both datasets to obtain the result of the rating. They include 1000 videos of 10 different categories and have implemented a web crawler to extract 1000 comments in each video. The experiments showed that the proposed approach achieved an accuracy of 75.43% on science and technology videos.

Ezpeleta *et al.* [30] proposed a method for filtering spam comments. In order to achieve that, they focused on sentiment analysis features and personality recognition features that assisting in detecting spam in social media platforms and classified text spam. They utilised a dataset

contains 6,431,471 comments and spam comments 481,334. Additionally, they a dataset involves 1000 spam and 300 ham was added to seven different classifiers to have the model built. Four datasets were applied, including an original dataset, polarity dataset, personality dataset, and combined dataset. As a result, they have achieved 82.55% accuracy using the model on the combined dataset.

TABLE I. THE PREVIOUS STUDIES

Author (s)	Approach	Dataset	Language
[19]	NB,SVM, and NBSVM	YouTube API	Indonesian
[2]	SVM	YouTube	Indonesian
[18]	RF ,SVM, and NB	YouTube Website	Indonesian and English
[4]	NB, RNNs, DT, NNs, RF and a hybrid model	Tweets API through GitHub, Kaggle and UCI.	English
[9]	NB	YouTube Selenium WebDriver	English
[20]	LR, RF, Linear Regressor, and RANSAC	Kaggle	English and Roman Urdu
[23]	LSTM, CNN, MLP, and CNN-LSTM	IMDB movie review.	English
[21]	CNN	YouTube API	English
[22]	NN	Twitter API	Korean and English
[25]	MLPNN, CNN, and RNN - LSTM	YouTube API	English
[29]	NLP, SentiStrength	Web API via HTTP	English

VI. CONCLUSION

This paper presents the related researches in the area of sentiment analysis for social media. It is clear that sentiment analysis is still an evolving field and needs to be searched deeply. Though researchers have come up with ways of sentiment analysis with Algorithms, which still has been the number of these studies was little in social media on YouTube. Sentiment analysis has been used in different fields of social media. and it is clear that sentiment analysis is still an evolving field. Regarding future work, we aim aimed to retrieve YouTube videos for children based on sentiment analysis classify to find out the content a useful or harmful using different types of Machine Learning (ML) classifications such as RF, LR, MNB, DT, KNN, AdaBoost, SVM, RNN, LSTM, CNN, CNN-LSTM and LSTM-CNN.

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